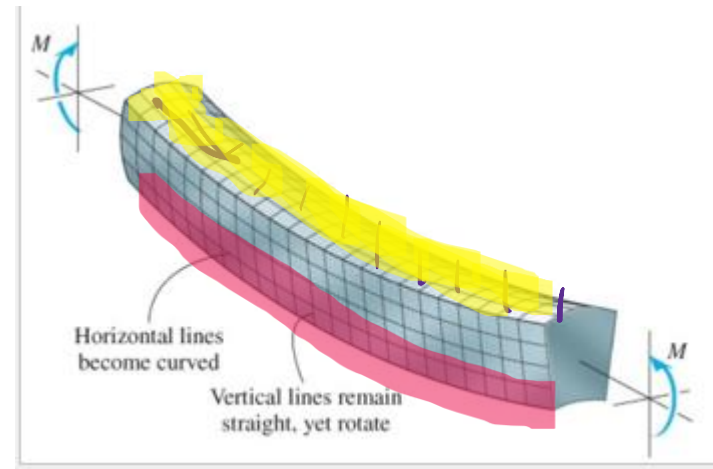
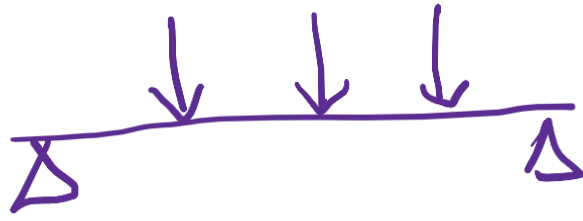
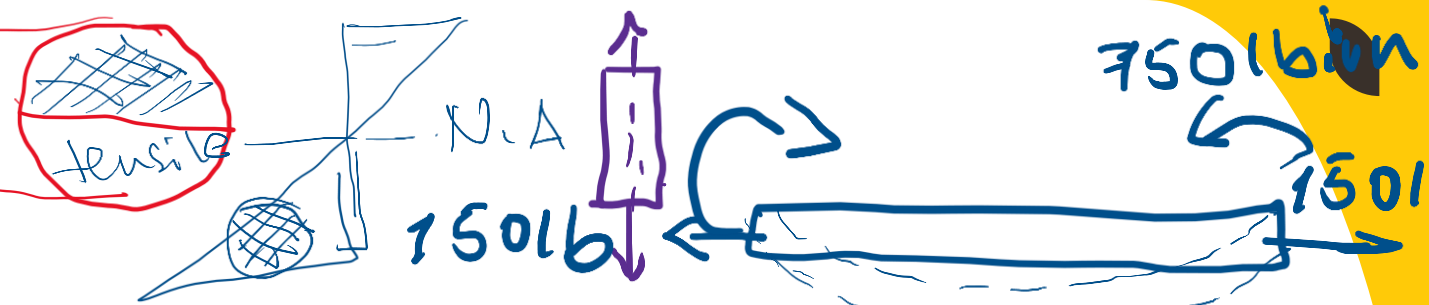
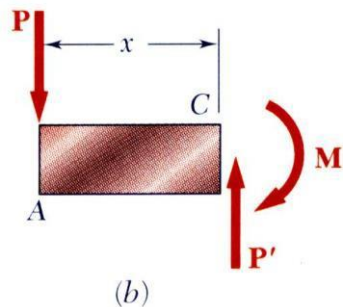
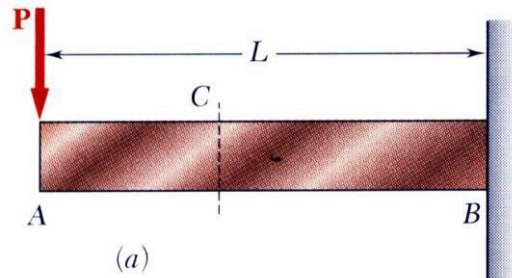
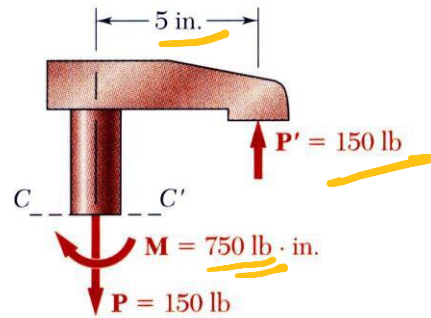
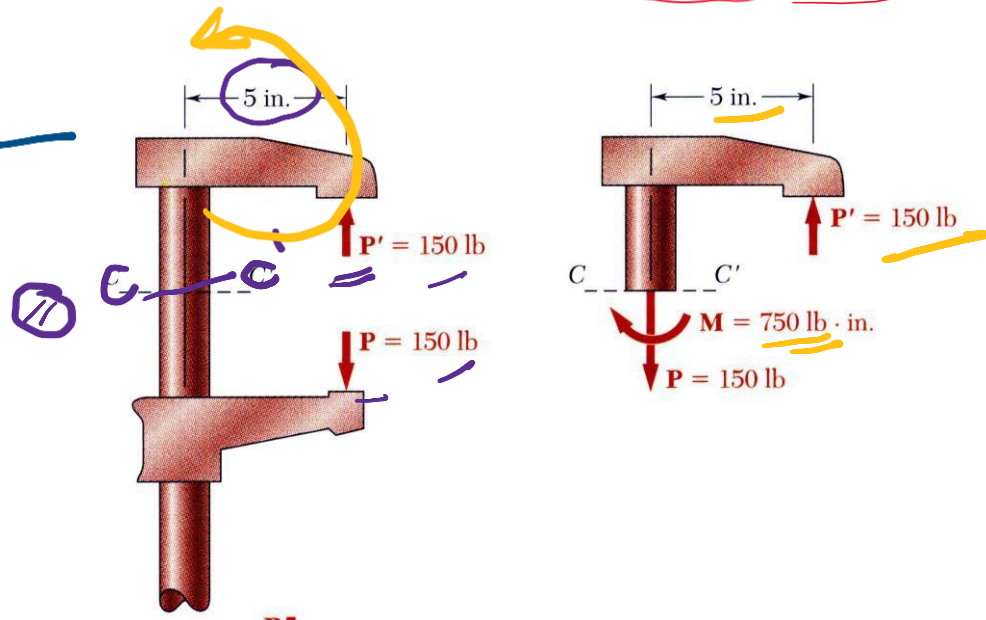




Chapter:

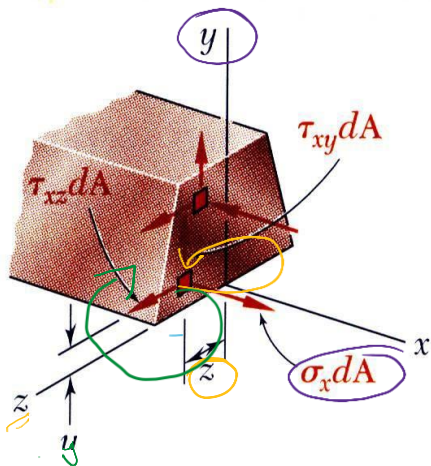
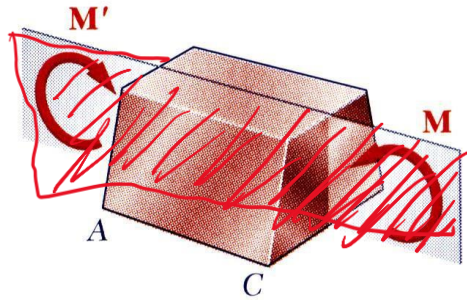
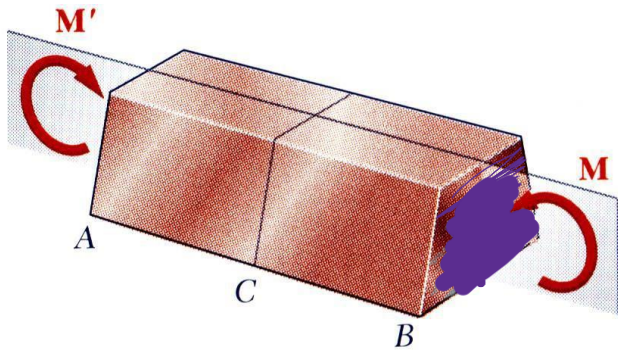
Stress in Beams

Other Loading Types



- *Eccentric Loading*: Axial loading which does not pass through section centroid produces internal forces equivalent to an axial force and a couple
- *Transverse Loading*: Concentrated or distributed transverse load produces internal forces equivalent to a shear force and a couple
- *Principle of Superposition*: The normal stress due to pure bending may be combined with the normal stress due to axial loading and shear stress due to shear loading to find the complete state of stress.

Symmetric Member in Pure Bending



- Internal forces in any cross section are equivalent to a couple. The moment of the couple is the section bending moment.
- From statics, a couple \$M\$ consists of two equal and opposite forces. *
- The sum of the components of the forces in any direction is zero.
- The moment is the same about any axis perpendicular to the plane of the couple and zero about any axis contained in the plane.
- These requirements may be applied to the sums of the components and moments of the statically indeterminate elementary internal forces.

$$M = F \cdot d_{\perp}$$

$$\delta_x = \frac{df}{dA}$$

$$\int df = \int \delta_x dA$$

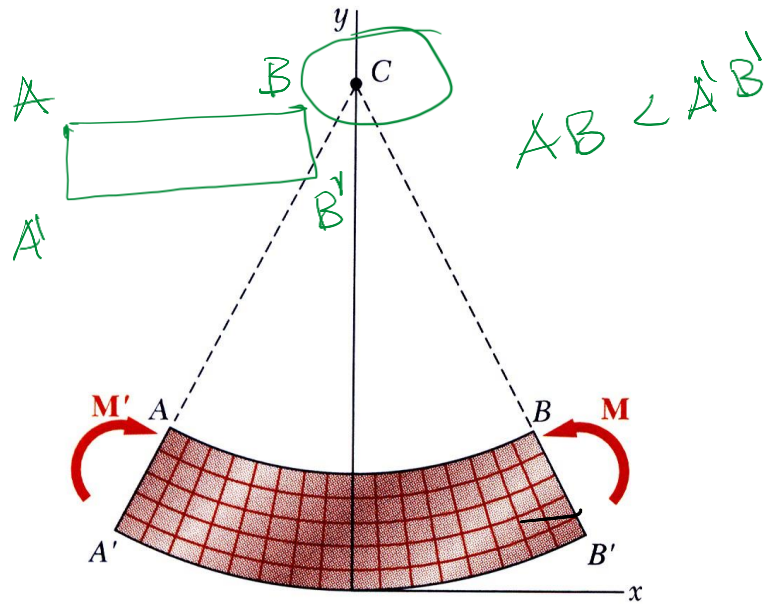
$$F = \int \delta_x dA$$

$$\underline{F_x} = \int \sigma_x dA = 0$$

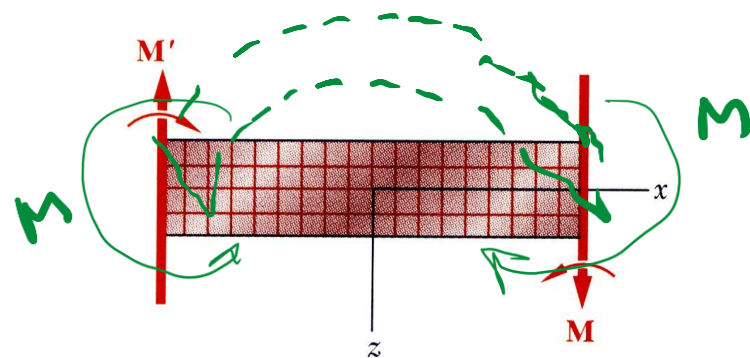
$$\underline{M_y} = \int z \sigma_x dA = 0$$

$$\underline{M_z} = \int -y \sigma_x dA = M$$

Bending Deformations



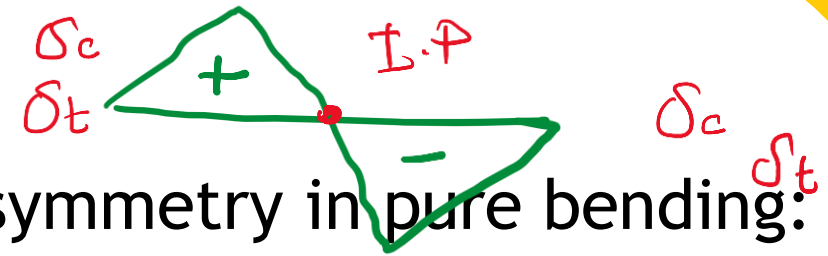
(a) Longitudinal, vertical section
(plane of symmetry)



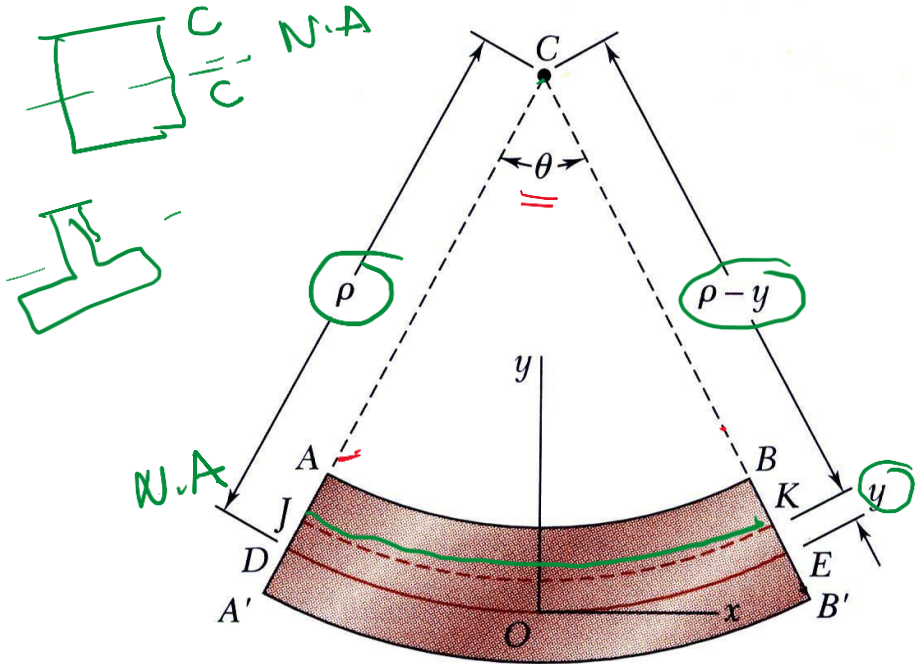
(b) Longitudinal, horizontal section

Beam with a plane of symmetry in pure bending:

- member remains symmetric
- bends uniformly to form a circular arc
- cross-sectional plane passes through arc center and remains planar
- length of top decreases and length of bottom increases
- a *neutral surface* must exist that is parallel to the upper and lower surfaces and for which the length does not change
- stresses and strains are negative (compressive) above the neutral plane and positive (tension) below it



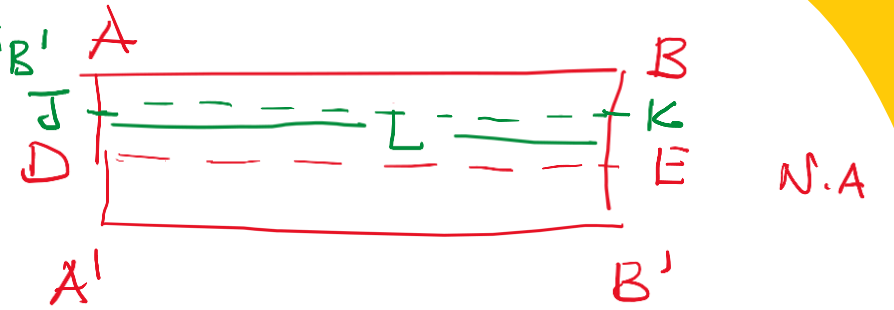
Strain Due to Bending



$$L = AB = DE = A'B'$$

$$l = DE = N.A$$

$$AB < A'B'$$



Consider a beam segment of length L .

After deformation, the length of the neutral surface remains L . At other sections,

$$\epsilon = \frac{\delta}{L_i} = \frac{L_f - L_i}{L_i}$$

$$L' = (\rho - y)\theta$$

$$\delta = L' - L = (\rho - y)\theta - \rho\theta = -y\theta$$

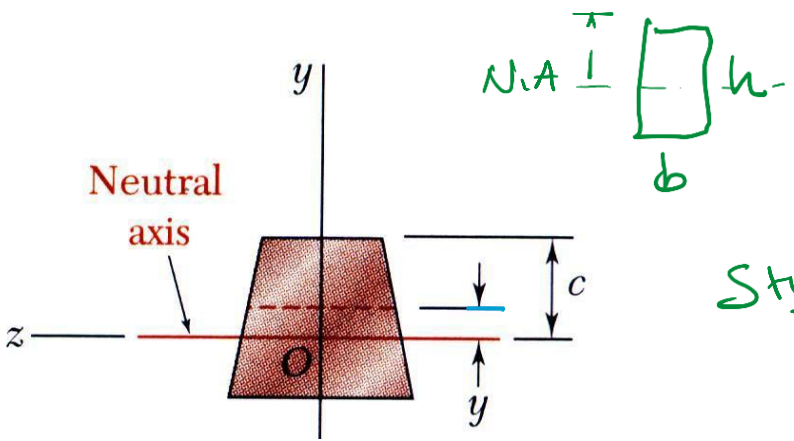
$$\epsilon_x = \frac{\delta}{L} = -\frac{y\theta}{\rho\theta} = -\frac{y}{\rho}$$

(strain varies linearly) With y

$$\epsilon_m = \frac{c}{\rho} \quad \text{or} \quad \rho = \frac{c}{\epsilon_m}$$

Strain Maximum

$$\epsilon_x = -\frac{y}{c} \epsilon_m$$



Stress Due to Bending

- For a linearly elastic material,

$$\sigma_x = E \epsilon_x = -\frac{y}{c} E \epsilon_m$$

$$\sigma_x = -\frac{y}{c} \sigma_m \quad (\text{stress varies linearly})$$

- For static equilibrium,

$$F_x = 0 = \int \sigma_x dA = \int -\frac{y}{c} \sigma_m dA$$

$$0 = -\frac{\sigma_m}{c} \int y dA$$

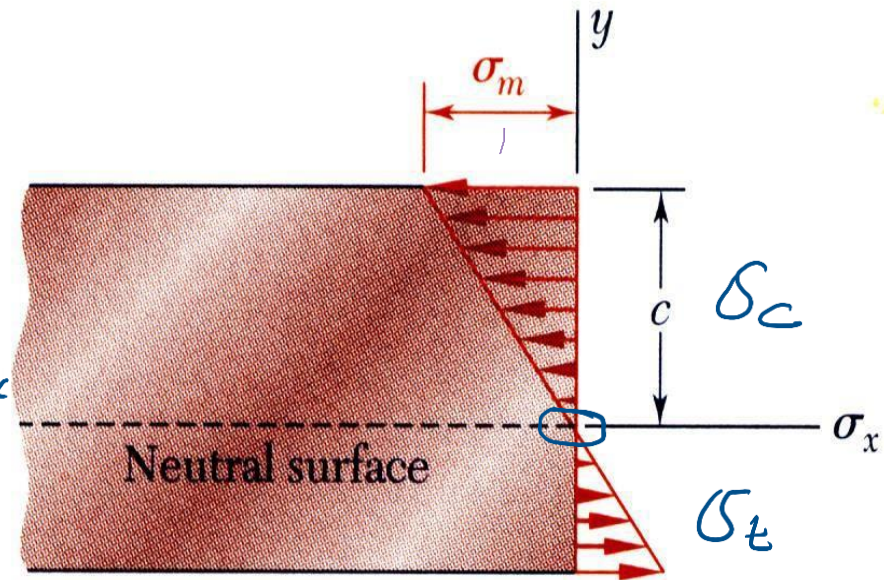
First moment of Area

First moment with respect to neutral plane is zero. Therefore, the neutral surface must pass through the section centroid.

$$\epsilon_x = -\frac{y}{c} \epsilon_m$$

$$E \cdot \epsilon_m = \sigma_{max}$$

$$y = \frac{M}{E} x$$



- For static equilibrium,

$$M_z = \int -y \sigma_x dA = \int -y \left(-\frac{y}{c} \sigma_m \right) dA$$

$$M = \frac{\sigma_m}{c} \int y^2 dA = \frac{\sigma_m I}{c}$$

$$\sigma_m = \frac{Mc}{I} = \frac{M}{S}$$

Second moment of Area (Moment of Inertia)

$$\sigma_m = \frac{\sigma_x \cdot c}{-\frac{y}{c} S} = \frac{c}{I}$$

Substituting $\sigma_x = -\frac{y}{c} \sigma_m$

$$\sigma_x = -\frac{My}{I}$$

Flexural Stress

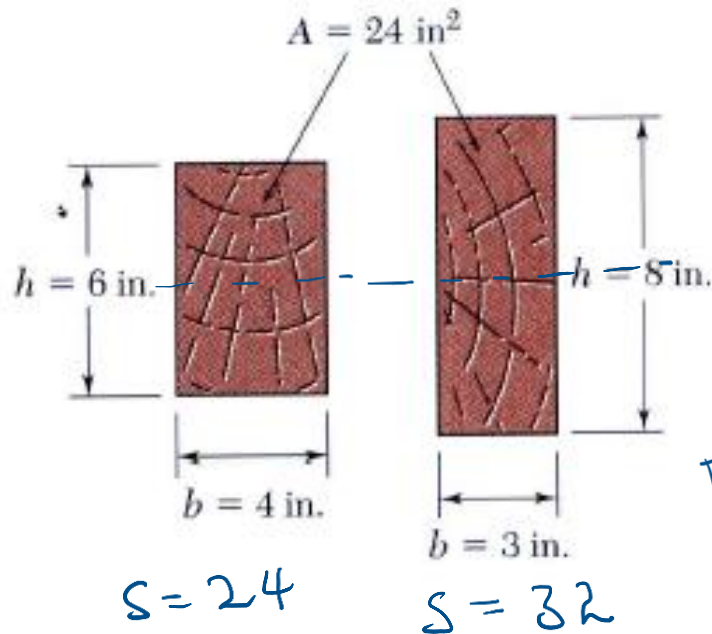
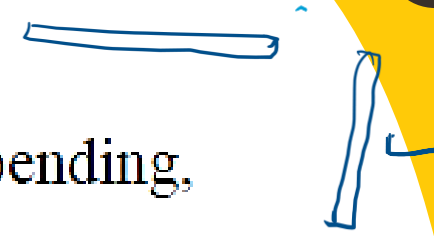
Flexural Formula

$$S = \frac{I}{c}$$

Section Modulus

Beam Section Properties

$$\sigma_m = \frac{M}{S}$$



- The maximum normal stress due to bending,

$$\sigma_m = \frac{Mc}{I} = \frac{M}{\underline{S}}$$

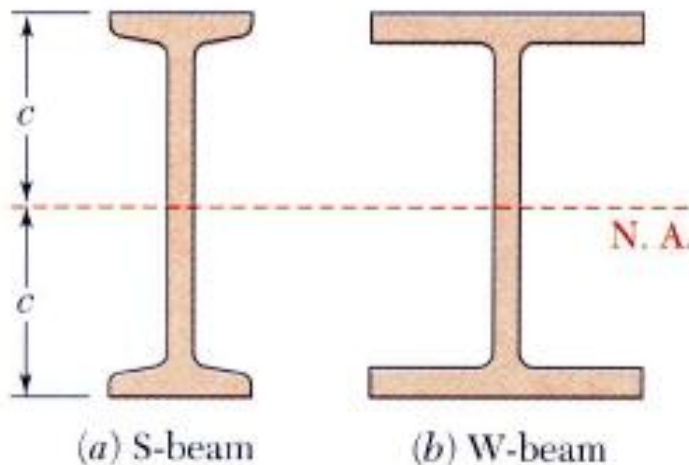
I = section moment of inertia

$$S = \frac{I}{c} = \text{section modulus}$$

$$S = \frac{I}{c}$$

$$\sigma_{all} = \frac{M}{S}$$

A beam section with a larger section modulus will have a lower maximum stress



- Consider a rectangular beam cross section,
Between two beams with the same cross sectional area, the beam with the greater depth will be more effective in resisting bending.
- Structural steel beams are designed to have a large section modulus.

Properties of American Standard Shapes

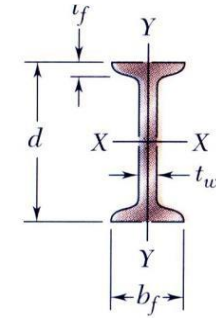
750 kg



Appendix C. Properties of Rolled-Steel Shapes (SI Units)

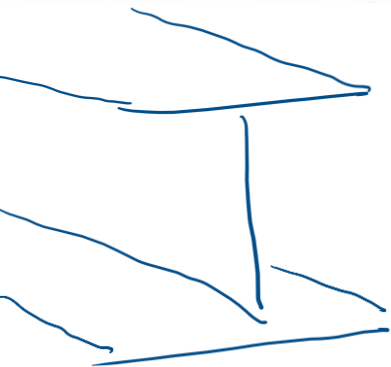
S Shapes (American Standard Shapes)

S 610 x 134



133

Designation†	Area A , mm ²	Depth d , mm	Flange		Web Thick- ness t_w , mm	Axis X-X			Axis Y-Y		
			Width b_f , mm	Thick- ness t_f , mm		I_x 10 ⁶ mm ⁴	S_x 10 ³ mm ³	r_x mm	I_y 10 ⁶ mm ⁴	S_y 10 ³ mm ³	r_y mm
S610 × 180	22900	622	204	27.7	20.3	1320	4240	240	34.9	341	39.0
158	20100	622	200	27.7	15.7	1230	3950	247	32.5	321	39.9
149	19000	610	184	22.1	18.9	995	3260	229	20.2	215	32.3
134	17100	610	181	22.1	15.9	938	<u>3080</u>	234	19.0	206	33.0
119	15200	610	178	22.1	12.7	878	2880	240	17.9	198	34.0
S510 × 143	18200	516	183	23.4	20.3	700	2710	196	21.3	228	33.9
128	16400	516	179	23.4	16.8	658	2550	200	19.7	216	34.4
112	14200	508	162	20.2	16.1	530	2090	193	12.6	152	29.5
98.3	12500	508	159	20.2	12.8	495	1950	199	11.8	145	30.4
S460 × 104	13300	457	159	17.6	18.1	385	1685	170	10.4	127	27.5
81.4	10400	457	152	17.6	11.7	333	1460	179	8.83	113	28.8
S380 × 74	9500	381	143	15.6	14.0	201	1060	145	6.65	90.8	26.1
64	8150	381	140	15.8	10.4	185	971	151	6.15	85.7	27.1



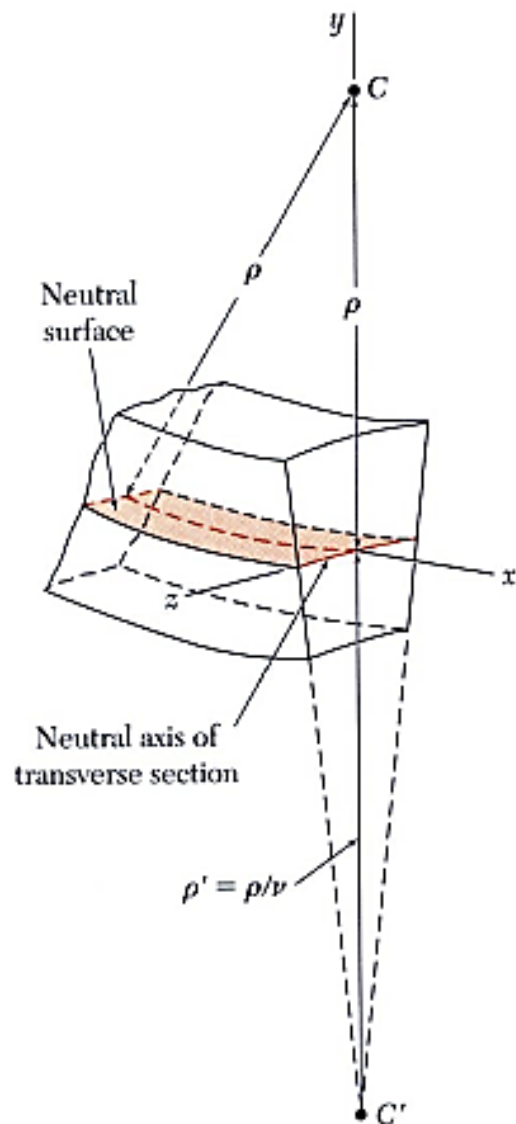
Flexural Formula(Bending Equation)

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y}$$


Where:

- M: Moment of Resistance ✓
 - I: Moment of Inertia of the section about the N.A. ✓
 - E: Youngs Modulus of Elasticity ✓
 - R: Radius of Curvature of N.A. ✓
 - σ : Bending Stress ✓
 - Y: Distance of a Point from the N.A. ✓
- The flexure formula gives the internal bending stress caused by an external moment on a beam or other bending member of homogeneous material.
 - It is derived here for a rectangular beam but is valid for any shape.

Deformations in a Transverse Cross Section



- Deformation due to bending moment M is quantified by the curvature of the neutral surface

$$\frac{1}{\rho} = \frac{\epsilon_m}{c} = \frac{\sigma_m}{Ec} = \frac{1}{Ec} \frac{Mc}{I} = \frac{M}{EI}$$

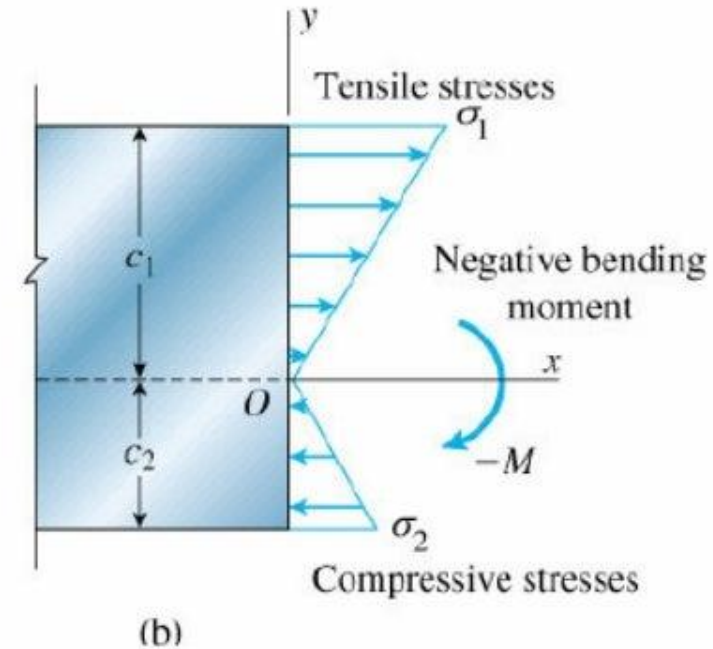
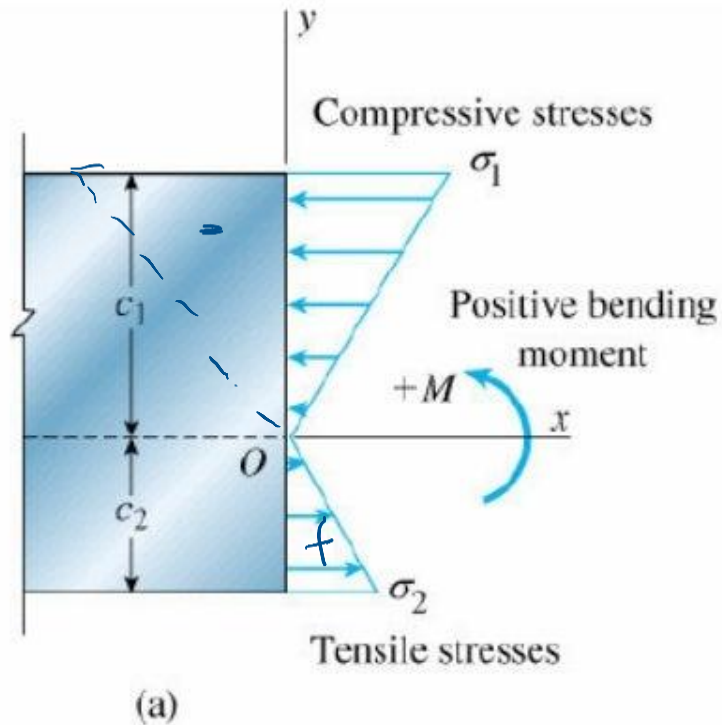
- Although cross sectional planes remain planar when subjected to bending moments, in-plane deformations are nonzero,

$$\epsilon_y = -\nu \epsilon_x = \frac{\nu y}{\rho} \quad \epsilon_z = -\nu \epsilon_x = \frac{\nu y}{\rho}$$

- Expansion above the neutral surface and contraction below it cause an in-plane curvature,

$$\frac{1}{\rho'} = \frac{\nu}{\rho} = \text{anticlastic curvature}$$

The maximum tensile and compressive stresses occur at the points located farthest from the N.A.



The maximal stresses;

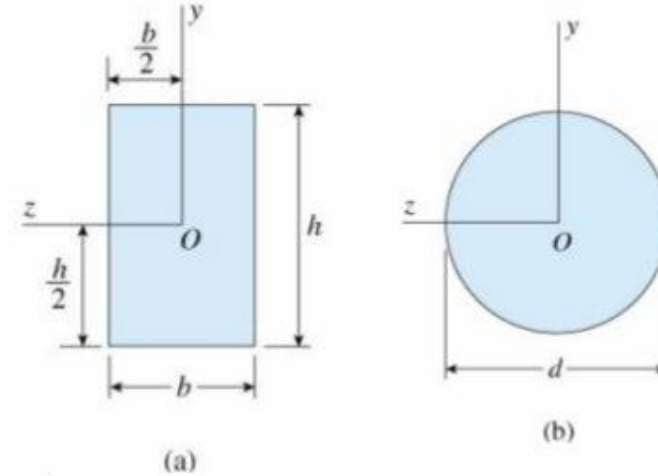
$$\sigma_1 = -\frac{M c_1}{I} = -\frac{M}{S_1}$$

$$\sigma_2 = \frac{M c_2}{I} = \frac{M}{S_2}$$

where $S_1 = \frac{I}{c_1}$,

$$S_2 = \frac{I}{c_2}$$

S_1 and **S_2** are known as the section moduli



for rectangular cross section

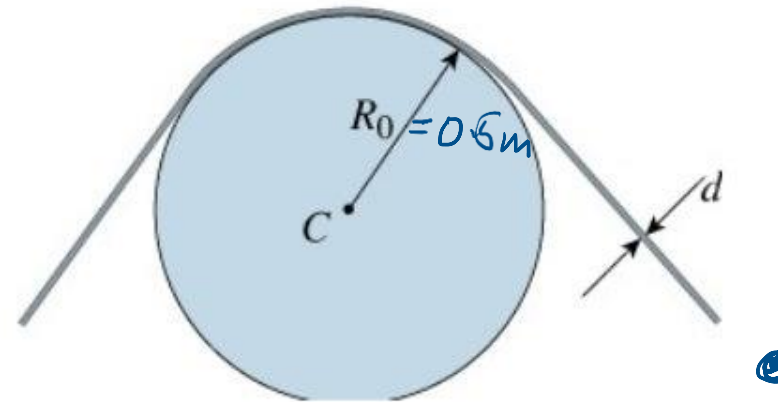
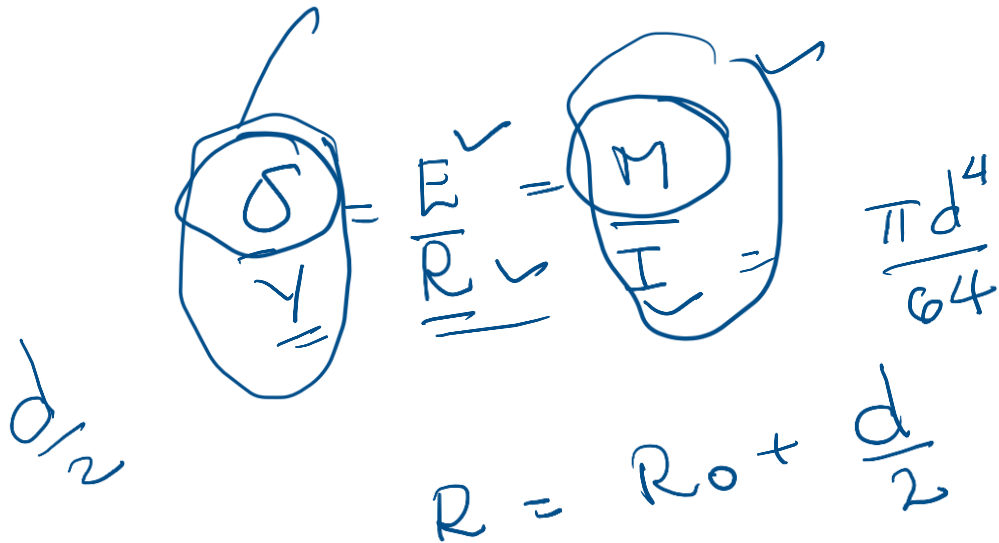
$$I = \frac{b h^3}{12} \quad S = \frac{b h^2}{6}$$

for circular cross section

$$I = \frac{\pi d^4}{64} \quad S = \frac{\pi d^3}{32}$$

Example

A steel wire of diameter $d = 4$ mm is bent around a cylindrical drum of radius $R_0 = 0.5$ m, $E = 200$ Gpa. Determine M and σ_{max} .





Example

Solution

The radius of curvature of the wire is;

$$\rho = R_0 + \frac{d}{2}$$

Bending moment;

$$\begin{aligned} M &= \frac{EI}{\rho} = \frac{2 EI}{2 R_0 + d} = \frac{\pi E d^4}{32(2R_0 + d)} \\ &= \frac{\pi (200 \times 10^3) 4^4}{32 (2 \times 500 + 4)} = 5007 \text{ N-mm} = \underline{\underline{5.007 \text{ N-m}}} \end{aligned}$$



The maximum bending stress;

$$\begin{aligned}\sigma_{max} &= \frac{M}{S} = \frac{M}{I / (d/2)} = \frac{M d}{2 I} \\ &= \frac{2 E I d}{2 I (2 R_0 + d)} = \frac{E d}{2 R_0 + d} \\ &= \frac{200 \times 10^3 \times 4}{2 \times 500 + 4} = 796.8 \text{ MPa}\end{aligned}$$

$$S = \frac{\pi d^3}{32}$$

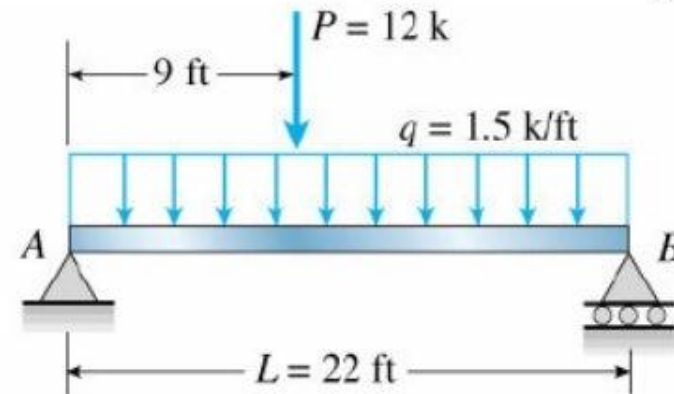
Example

A simple beam AB of length $L = 6.7$ m

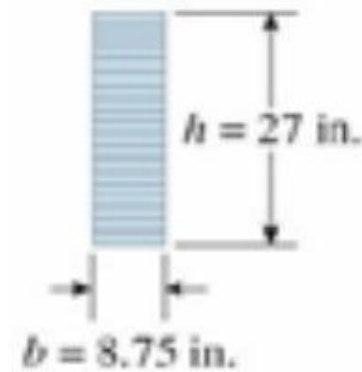
$$q = 22\text{kN/m}, P = 50\text{kN}$$

$$b = 220\text{mm}, h = 700\text{mm}$$

Determine the maximum tensile and compressive stresses due to bending.



(a)



(b)



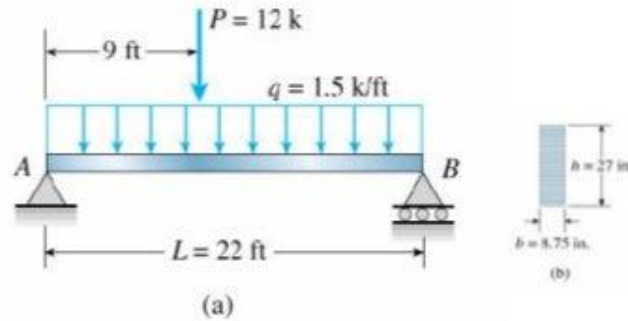
Example

Solution

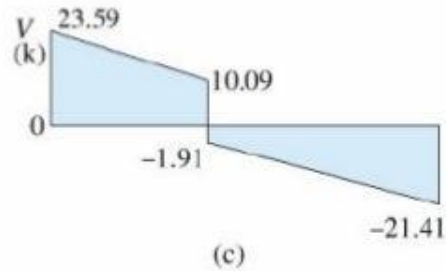
- 1) Construct the M-diagram and V-diagram.
- 2) Find the max. bending moment along the beam.
- 3) Calculate the max. bending stresses.



M- and V-diagram.

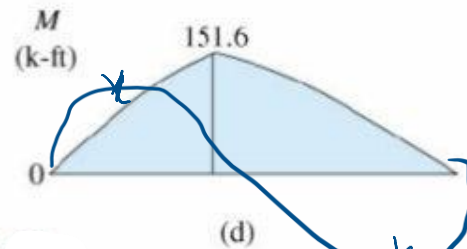


Max. bending stress occurs at the section of $M_{\max}$.



$$M_{\max} = \underline{\underline{193.9 \text{ kN-m}}}$$

σ_c
 σ_t



σ_c
 σ_t



The section modulus **S** is;

$$S = \frac{b h^2}{6} = \frac{0.22 \times 0.7^2}{6} = 0.018 \text{ m}^3$$



700

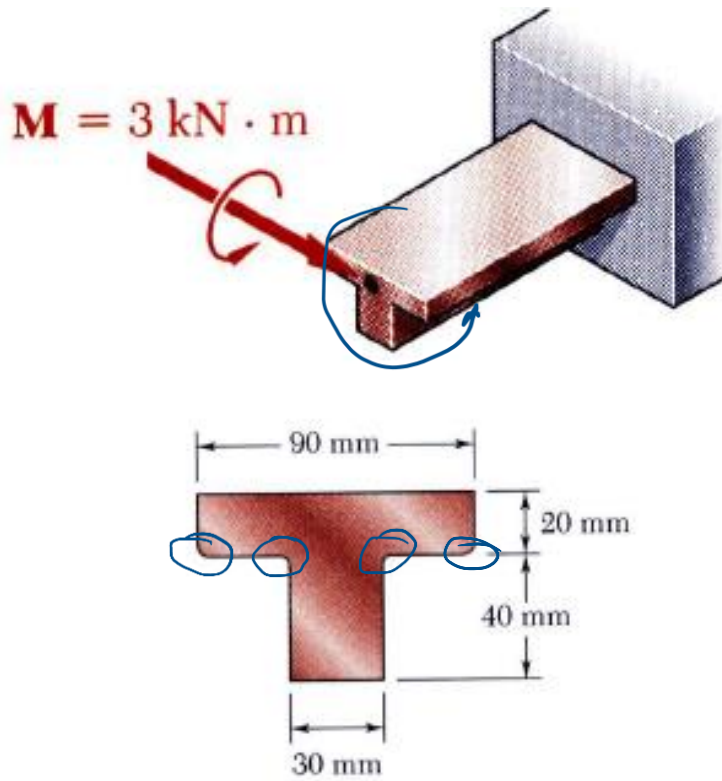
220

The max. bending stresses are;

$$\underline{\sigma_t} = \sigma_2 = \frac{M}{S} = \frac{139.9 \text{ kN-m}}{0.018 \text{ m}^3} = \underline{\underline{10.8 \text{ MPa}}}$$

$$\underline{\sigma_c} = \sigma_1 = -\frac{M}{S} = -\underline{\underline{10.8 \text{ MPa}}}$$

Example



A cast-iron machine part is acted upon by a 3 kN-m couple. Knowing $E = 165$ GPa and neglecting the effects of fillets, determine (a) the maximum tensile and compressive stresses, (b) the radius of curvature.

SOLUTION:

- Based on the cross section geometry, calculate the location of the section centroid and moment of inertia.

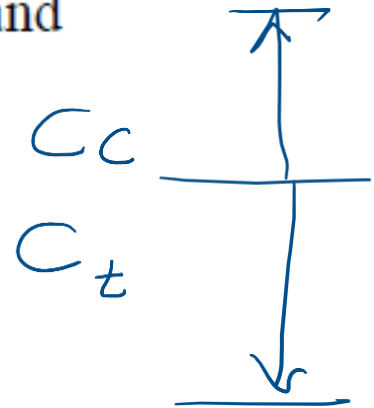
$$\bar{Y} = \frac{\sum \bar{y}A}{\sum A} \quad I_{x'} = \sum (\bar{I} + Ad^2)$$

- Apply the elastic flexural formula to find the maximum tensile and compressive stresses.

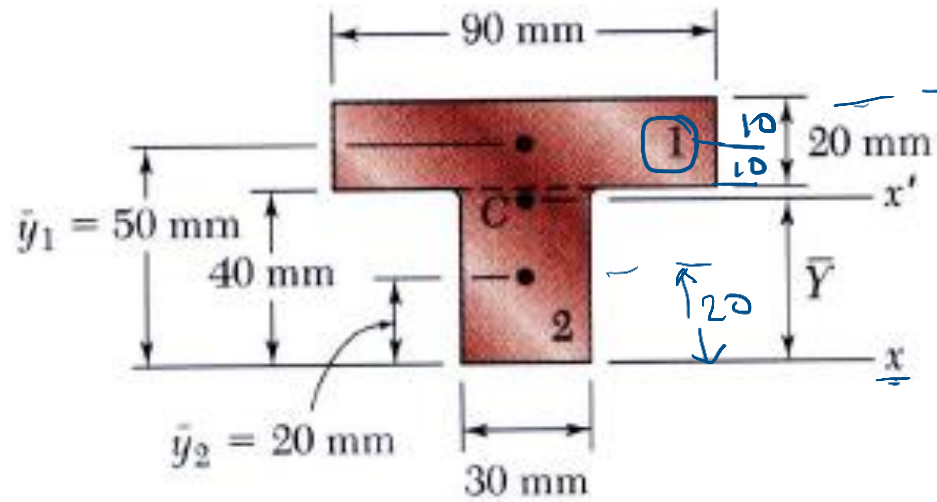
$$\sigma_m = \frac{Mc}{I}$$

- Calculate the curvature

$$\frac{E}{\rho} = \frac{M}{I}$$



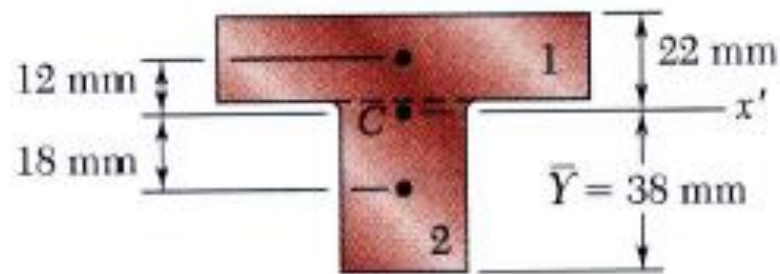
SOLUTION:



Based on the cross section geometry, calculate the location of the section centroid and moment of inertia.

	Area, mm ²	$\bar{y}$, mm	$\bar{y}A$, mm ³
1	$20 \times 90 = 1800$	10 50	90×10^3
2	$40 \times 30 = 1200$	40 20	24×10^3
	$\Sigma A = \underline{\underline{3000}}$		$\Sigma \bar{y}A = \underline{\underline{114 \times 10^3}}$

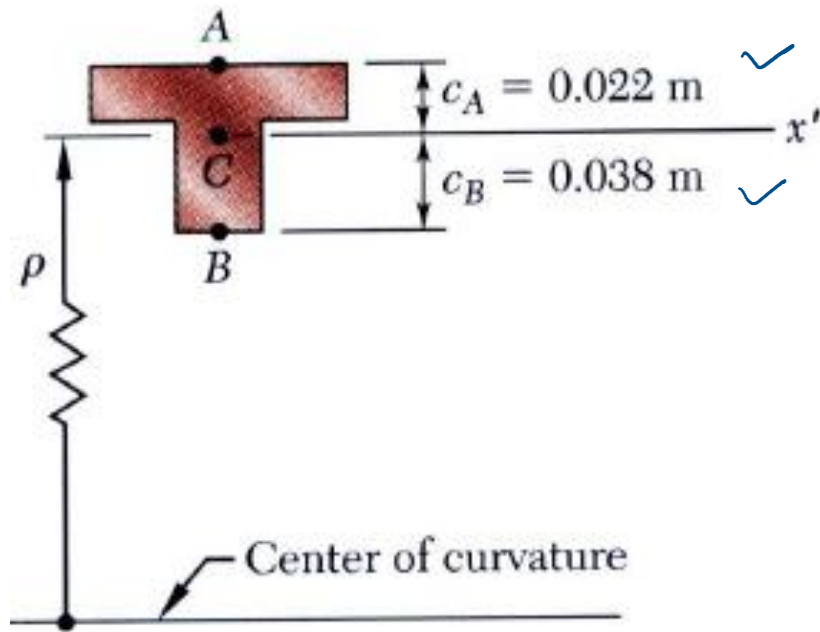
$$\bar{Y} = \frac{\Sigma \bar{y}A}{\Sigma A} = \frac{114 \times 10^3}{3000} = \underline{\underline{38 \text{ mm}}} \quad 22 \text{ mm}$$



$$I_{x'} = \Sigma (\bar{I} + Ad^2) = \Sigma \left(\frac{1}{12}bh^3 + Ad^2 \right)$$

$$= \left(\frac{1}{12}90 \times 20^3 + 1800 \times 12^2 \right) + \left(\frac{1}{12}30 \times 40^3 + 1200 \times 18^2 \right)$$

$$I = 868 \times 10^3 \text{ mm}^4 = 868 \times 10^{-9} \text{ m}^4$$



- Apply the elastic flexural formula to find the maximum tensile and compressive stresses.

$$\sigma_m = \frac{Mc}{I}$$

$$\sigma_A = \frac{M c_A}{I} = \frac{3 \text{ kN} \cdot \text{m} \times 0.022 \text{ m}}{868 \times 10^{-9} \text{ mm}^4}$$

$$\sigma_A = +76.0 \text{ MPa}$$

σ_t

$$\sigma_B = -\frac{M c_B}{I} = -\frac{3 \text{ kN} \cdot \text{m} \times 0.038 \text{ m}}{868 \times 10^{-9} \text{ mm}^4}$$

$$\sigma_B = -131.3 \text{ MPa}$$

σ_c

- Calculate the curvature

$$\frac{1}{\rho} = \frac{M}{EI}$$

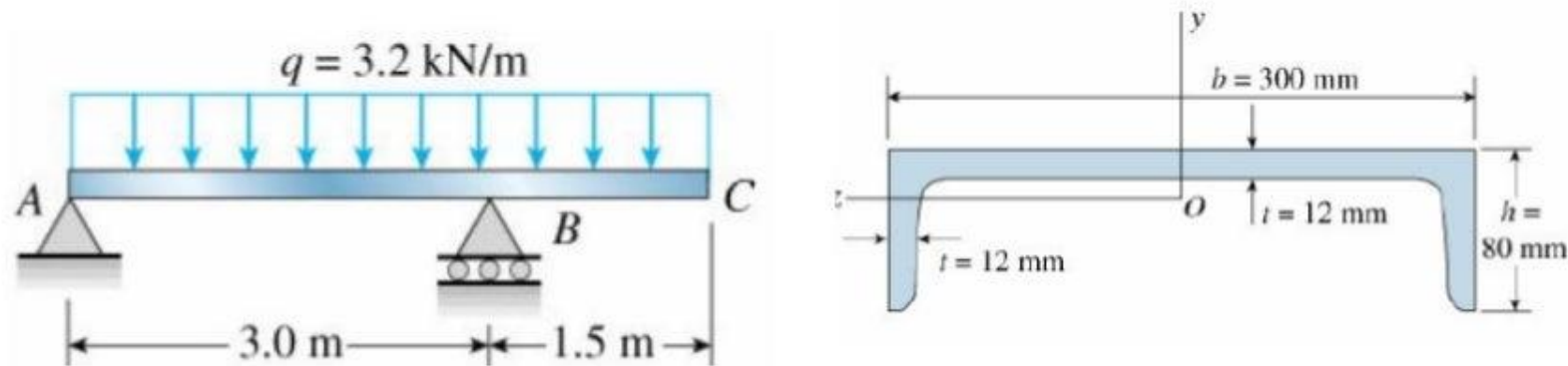
$$= \frac{3 \text{ kN} \cdot \text{m}}{(165 \text{ GPa})(868 \times 10^{-9} \text{ m}^4)}$$

$$\frac{1}{\rho} = 20.95 \times 10^{-3} \text{ m}^{-1}$$

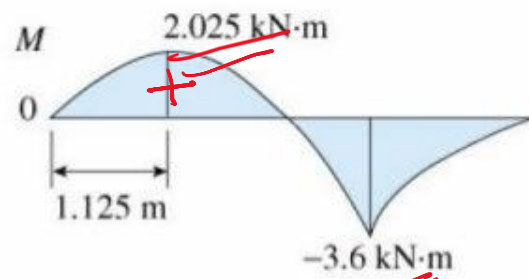
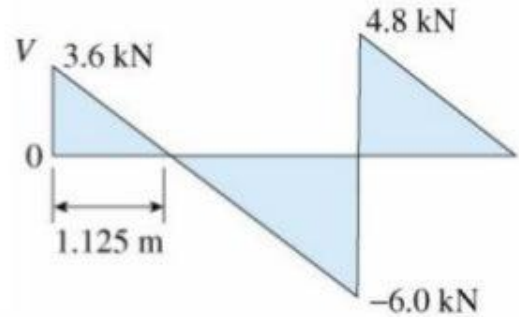
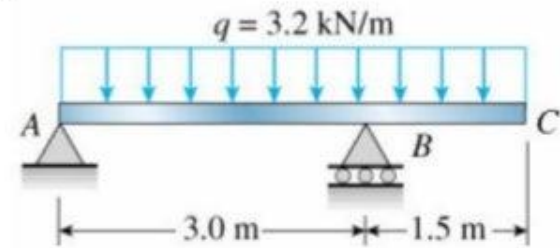
$$\rho = 47.7 \text{ m}$$

Example

An overhanged beam ABC subjected uniform load of intensity $q = 3.2 \text{ kN/m}$ for the cross section (channel section) $t = 12 \text{ mm}$, $b = 300 \text{ mm}$, $h = 80 \text{ mm}$. Determine the maximum tensile and compressive stresses in the beam.



✓
I

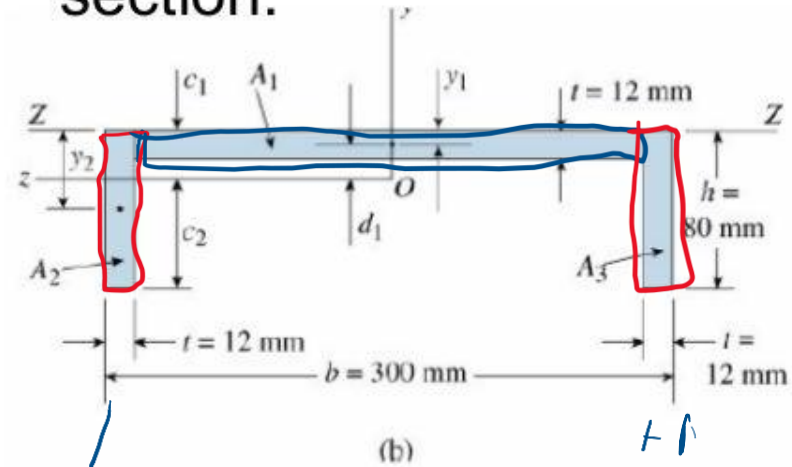


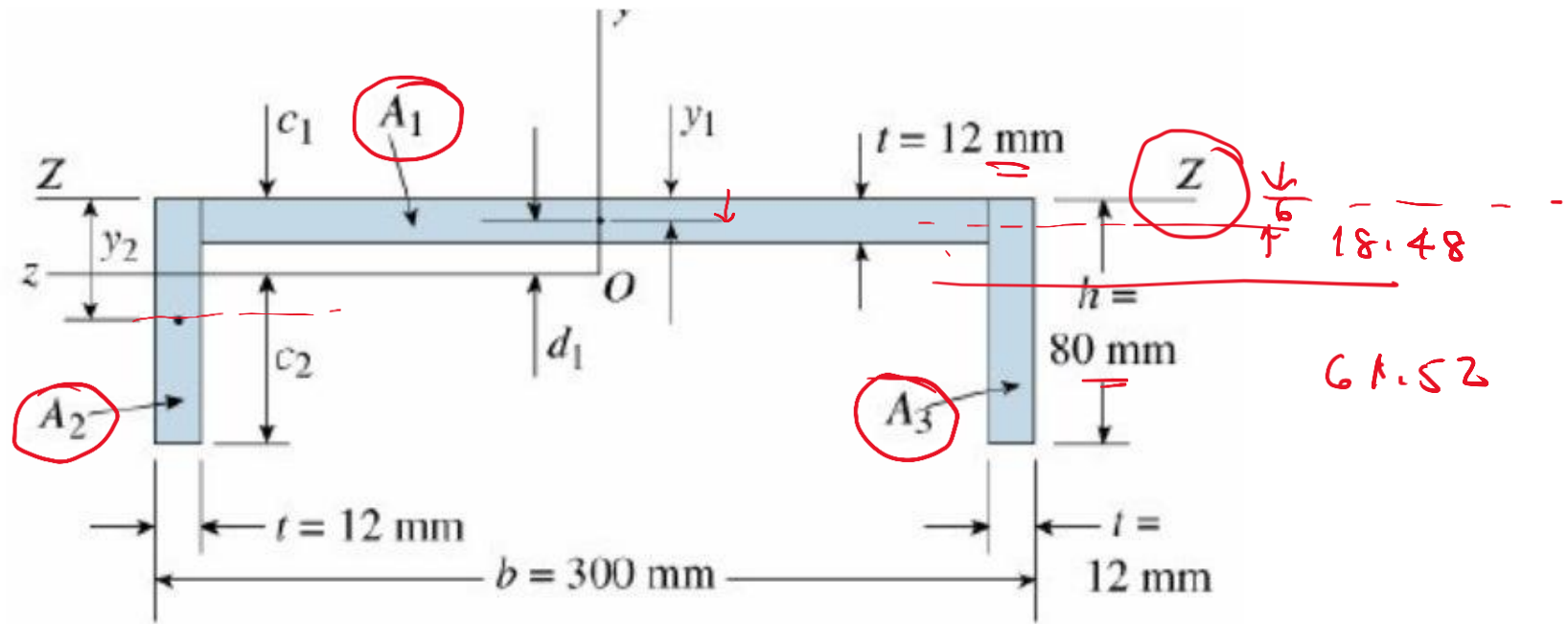
(c)

$$+M_{max} = 2.205 \text{ kN-m}$$

$$-M_{max} = -3.6 \text{ kN-m}$$

Find the N.A. of the section:





	$A(\text{mm}^2)$	$y(\text{mm})$	$A y (\text{mm}^3)$
A_1	3,312 ✓	6 ✓	19,872
A_2	960	40	38,400
A_3	960	40	38,400
total	5,232		96,672

$$\bar{Y} \quad c_1 = \frac{\sum A_i y_i}{\sum A_i} = \frac{96,672}{5,232} = \underline{\underline{18.48 \text{ mm}}}$$

$$\bar{Y} \quad c_2 = h - c_1 = \underline{\underline{61.52 \text{ mm}}}$$

Moment of inertia of the section is

$$I_{z1} = I_{zc} + A_1 d_1^2$$

$$I_{zc} = \frac{1}{12} (b - 2t) \underline{t^3} = \frac{1}{12} 276 \times 12^3 = \underline{39744} \text{ mm}^4$$

$$d_1 = c_1 - t/2 = 12.48 \text{ mm}$$

$$I_{z1} = \underline{39,744} + 3,312 \times 12.48^2 = 555,600 \text{ mm}^4$$

$$\text{similarly } \underline{I_{z2}} = \underline{I_{z3}} = \underline{956,000} \text{ mm}^4$$

Then the centroidal moment of inertia I_z is;

$$I_z = I_{z1} + I_{z2} + I_{z3} = 2.469 \times 10^6 \text{ mm}^4$$

$$C_1 = 18.48$$

$$C_2 = 61.52$$

$$S_2 = \frac{I}{C_2}$$

At the section of maximum positive moment;

$$\sigma_t = \sigma_2 = \frac{M}{S_2} = \frac{2.025 \times 10^3 \times 10^3}{40,100} = 50.5 \text{ MPa} \quad \sigma_t$$

$$\sigma_c = \sigma_1 = -\frac{M}{S_1} = -\frac{2.025 \times 10^3 \times 10^3}{133,600} = -15.2 \text{ MPa} \quad \sigma_c$$

At the section of maximum positive moment;

$$\sigma_t = \sigma_1 = -\frac{M}{S_1} = -\frac{-3.6 \times 10^3 \times 10^3}{133,600} = 26.9 \text{ MPa} \quad \sigma_t$$

$$\sigma_c = \sigma_2 = \frac{M}{S_2} = \frac{-3.6 \times 10^3 \times 10^3}{40,100} = -89.8 \text{ MPa} \quad \sigma_c$$



Thus $(\sigma_t)_{max}$ occurs at the section of maximum positive moment;

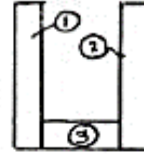
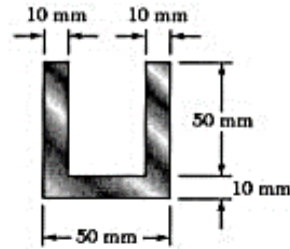
$$\underline{\underline{(\sigma_t)_{max} = 50.5 \text{ Mpa}}}$$

AND $(\sigma_c)_{max}$ occurs at the section of maximum negative moment;

$$\underline{\underline{(\sigma_c)_{max} = - 89.8 \text{ Mpa}}}$$

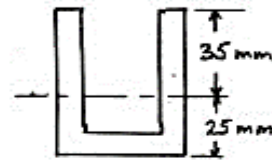
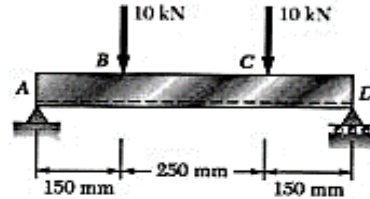
Problem 4.9

4.7 through 4.9 Two vertical forces are applied to a beam of the cross section shown. Determine the maximum tensile and compressive stresses in portion BC of the beam.



	A, mm^2	$\bar{y}_o, \text{mm}$	$A\bar{y}_o, \text{mm}^3$
①	600	30	18×10^3
②	600	30	18×10^3
③	300	5	1.5×10^3
	1500		37.5×10^3

$$\bar{y}_o = \frac{37.5 \times 10^3}{1500} = 25 \text{ mm}$$



Neutral axis lies 25 mm above the base.

$$I_1 = \frac{1}{12} (10)(60)^3 + (600)(5)^2 = 195 \times 10^3 \text{ mm}^4$$

$$I_2 = I_1 = 195 \text{ mm}^4$$

$$I_3 = \frac{1}{12} (30)(10)^3 + (300)(20)^2 = 122.5 \times 10^3 \text{ mm}^4$$

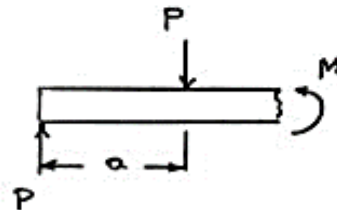
$$I = I_1 + I_2 + I_3 = 512.5 \times 10^3 \text{ mm}^4 = 512.5 \times 10^{-9} \text{ m}^4$$

$$y_{\text{top}} = 35 \text{ mm} = 0.035 \text{ m}$$

$$y_{\text{bot}} = -25 \text{ mm} = -0.025 \text{ m}$$

$$a = 150 \text{ mm} = 0.150 \text{ m} \quad P = 10 \times 10^3 \text{ N}$$

$$M = Pa = (10 \times 10^3)(0.150) = 1.5 \times 10^3 \text{ N}\cdot\text{m}$$



$$\sigma_{\text{top}} = - \frac{M y_{\text{top}}}{I} = - \frac{(1.5 \times 10^3)(0.035)}{512.5 \times 10^{-9}} = - 102.4 \times 10^6 \text{ Pa}$$

$$\sigma_{\text{top}} = -102.4 \text{ MPa (compression)}$$

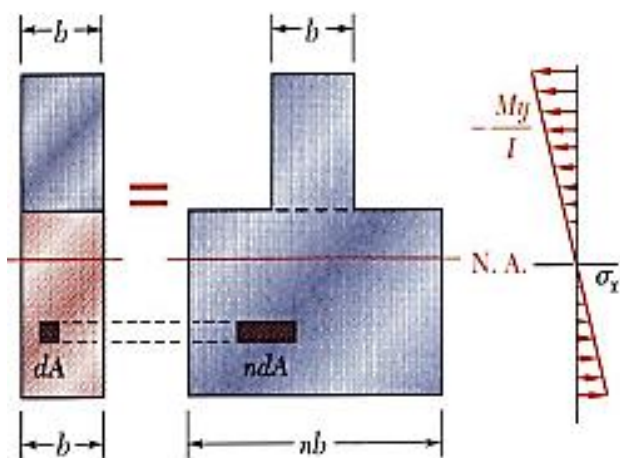
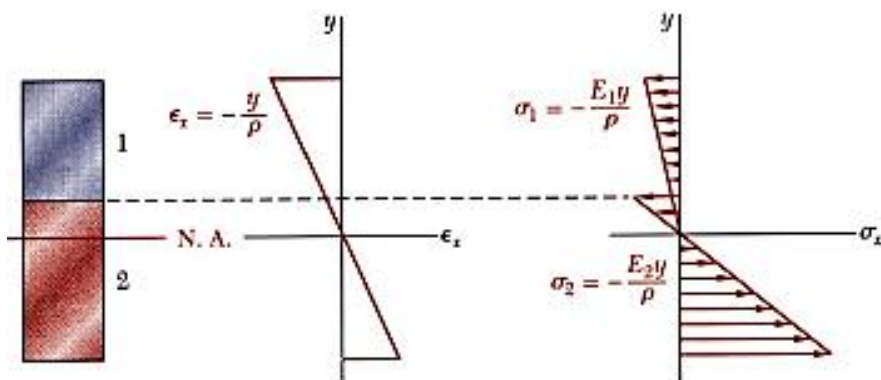
$$\sigma_{\text{bot}} = - \frac{M y_{\text{bot}}}{I} = - \frac{(1.5 \times 10^3)(-0.025)}{512.5 \times 10^{-9}} = 73.2 \times 10^6 \text{ Pa}$$

$$\sigma_{\text{bot}} = 73.2 \text{ MPa (tension)}$$

U c

v t

Bending of Members Made of Several Materials



$$\sigma_x = -\frac{My}{I}$$

$$\sigma_1 = \sigma_x \quad \sigma_2 = n\sigma_x$$

- Consider a composite beam formed from two materials with E_1 and E_2 .

- Normal strain varies linearly.

$$\epsilon_x = -\frac{y}{\rho}$$

- Piecewise linear normal stress variation.

$$\sigma_1 = E_1 \epsilon_x = -\frac{E_1 y}{\rho} \quad \sigma_2 = E_2 \epsilon_x = -\frac{E_2 y}{\rho}$$

Neutral axis does not pass through section centroid of composite section.

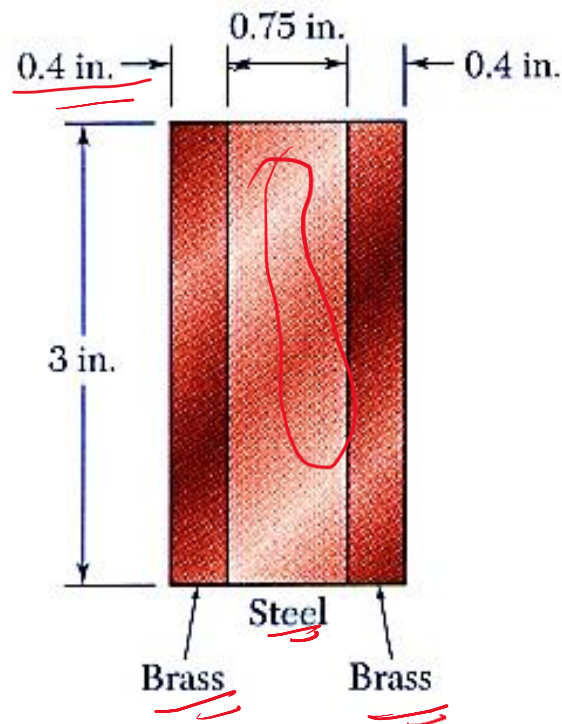
- Elemental forces on the section are

$$dF_1 = \sigma_1 dA = -\frac{E_1 y}{\rho} dA \quad dF_2 = \sigma_2 dA = -\frac{E_2 y}{\rho} dA$$

- Define a transformed section such that

$$dF_2 = -\frac{(nE_1)y}{\rho} dA = -\frac{E_1 y}{\rho} (n dA) \quad n = \frac{E_2}{E_1}$$

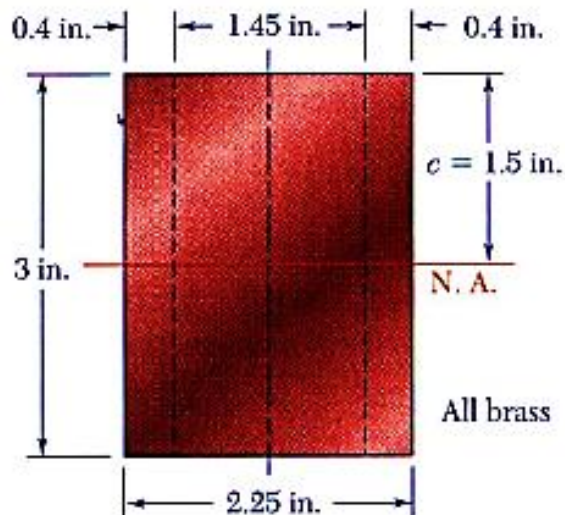
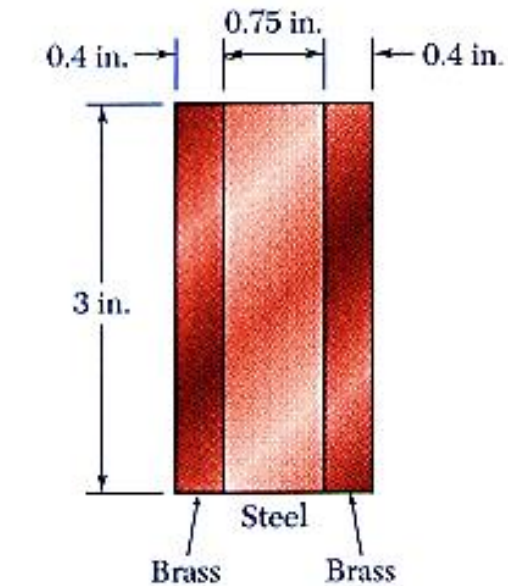
Example



Bar is made from bonded pieces of steel ($E_s = 29 \times 10^6$ psi) and brass ($E_b = 15 \times 10^6$ psi). Determine the maximum stress in the steel and brass when a moment of 40 kip*in is applied.

- Transform the bar to an equivalent cross section made entirely of brass
- Evaluate the cross sectional properties of the transformed section
- Calculate the maximum stress in the transformed section. This is the correct maximum stress for the brass pieces of the bar.
- Determine the maximum stress in the steel portion of the bar by multiplying the maximum stress for the transformed section by the ratio of the moduli of elasticity.

SOLUTION:



- Transform the bar to an equivalent cross section made entirely of brass.

$$n = \frac{E_s}{E_b} = \frac{29 \times 10^6 \text{ psi}}{15 \times 10^6 \text{ psi}} = 1.933$$

$$b_T = 0.4 \text{ in} + 1.933 \times 0.75 \text{ in} + 0.4 \text{ in} = 2.25 \text{ in}$$

- Evaluate the transformed cross sectional properties

$$I = \frac{1}{12} b_T h^3 = \frac{1}{12} (2.25 \text{ in.})(3 \text{ in})^3 = 5.063 \text{ in}^4$$

- Calculate the maximum stresses

$$\sigma_m = \frac{Mc}{I} = \frac{(40 \text{ kip} \cdot \text{in})(1.5 \text{ in})}{5.063 \text{ in}^4} = 11.85 \text{ ksi}$$

$$(\sigma_b)_{\max} = \sigma_m$$

$$(\sigma_s)_{\max} = n \sigma_m = 1.933 \times 11.85 \text{ ksi}$$

$$(\sigma_b)_{\max} = 11.85 \text{ ksi}$$

$$(\sigma_s)_{\max} = 22.9 \text{ ksi}$$

Stress Concentration

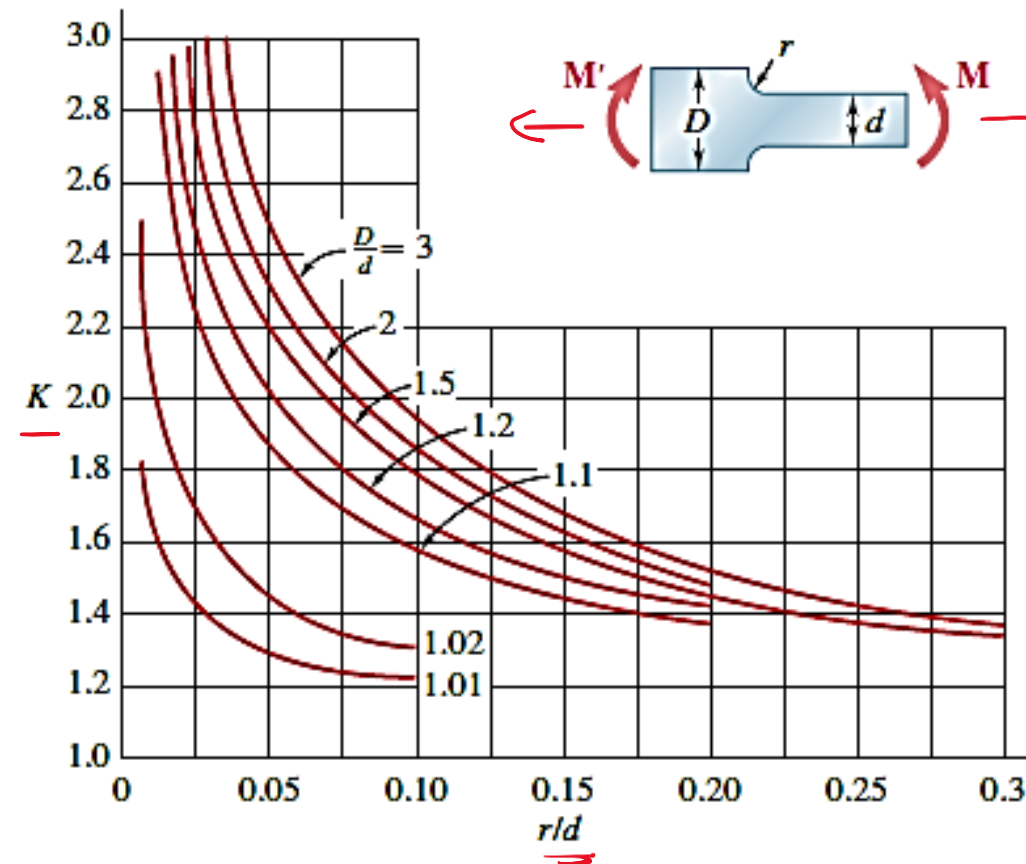


Fig. 4.24 Stress-concentration factors for *flat bars* with fillets under pure bending. (Source: W. D. Pilkey and D. F. Pilkey, *Peterson's Stress Concentration Factors*, 3rd ed., John Wiley & Sons, New York, 2008.)

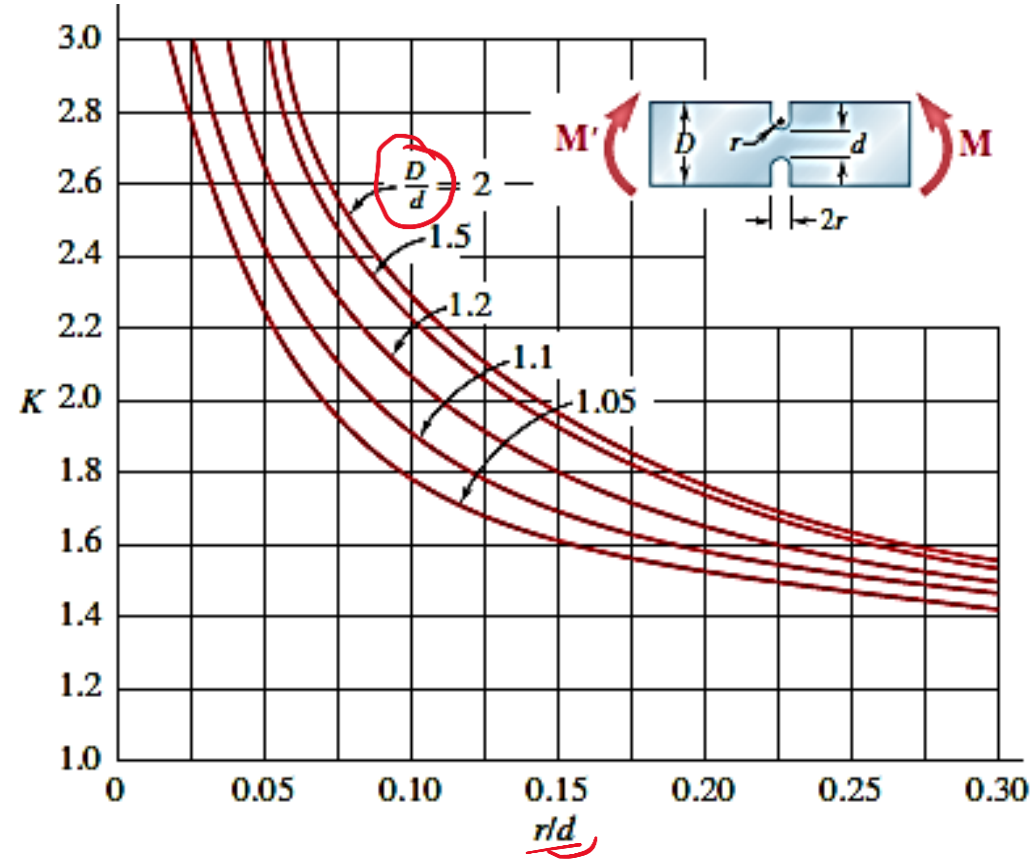
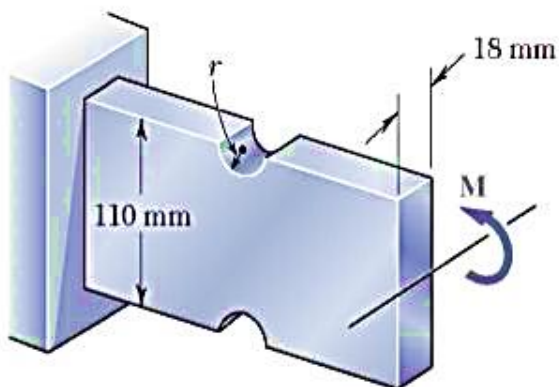


Fig. 4.25 Stress-concentration factors for *flat bars* with grooves (notches) under pure bending. (Source: W. D. Pilkey and D. F. Pilkey, *Peterson's Stress Concentration Factors*, 3rd ed., John Wiley & Sons, New York, 2008.)

$$\sigma_m = K \frac{Mc}{I}$$

$$\sigma_{max} = K \sigma_{nominal} \quad \sigma_{nominal} = \frac{Mc}{I}$$

Problem 4.61



4.61 Semicircular grooves of radius r must be milled as shown in the sides of a steel member. Using an allowable stress of 55 MPa, determine the largest bending moment that can be applied to the member when the radius r of the semicircular grooves is (a) $r = 10$ mm, (b) $r = 18$ mm.

$$(a) \quad d = D - 2r = 110 - 2(10) = 90 \text{ mm}$$

$$\frac{D}{d} = \frac{110}{90} = 1.22, \quad \frac{r}{d} = \frac{10}{90} = 0.09$$

From Fig. 4.32 $K = 2.16$.

$$I = \frac{1}{12} b h^3 = \frac{1}{12} (18) (90)^3 = 1093500 \text{ mm}^4 \quad c = 45 \text{ mm}$$

$$\sigma = K \frac{M c}{I} \therefore M = \frac{\sigma I}{K c} = \frac{(55 \times 10^6) (1.0935 \times 10^{-6})}{(2.16) (0.045)} = 618.8 \text{ Nm}$$

$$(b) \quad d = D - 2r = 110 - 2(18) = 74 \text{ mm}, \quad \frac{D}{d} = \frac{110}{74} = 1.5, \quad \frac{r}{d} = \frac{18}{74} = 0.24$$

$$\text{From Fig. 4.32 } K = 1.62, \quad I = \frac{1}{12} b h^3 = \frac{1}{12} (18) (74)^3 = 607836 \text{ mm}^4$$

$$c = \frac{1}{2} d = 37 \text{ mm} \quad M = \frac{\sigma I}{K c} = \frac{(55 \times 10^6) (607836 \times 10^{-12})}{(1.62) (0.037)} = 557.7 \text{ Nm}$$



Photo 4.6 Bench press.

Eccentric Axial Loading in a Plane Symmetry

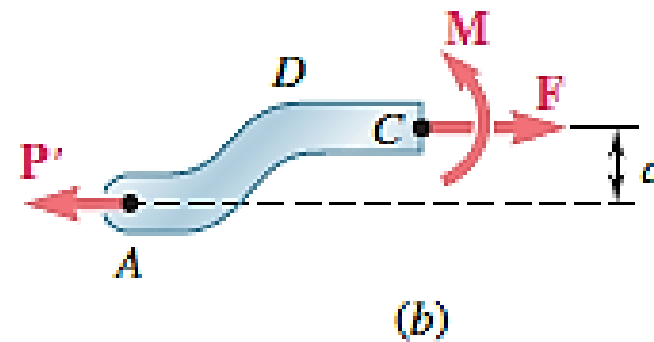
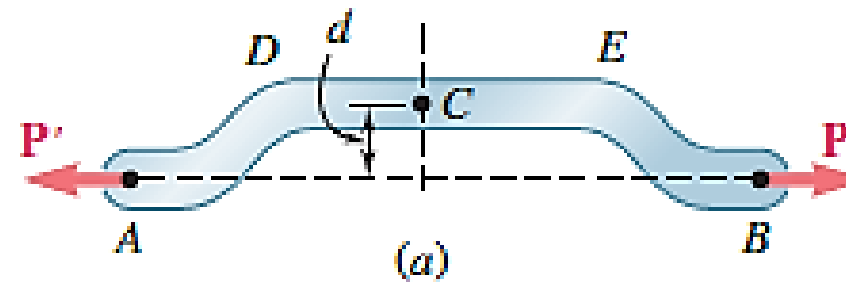
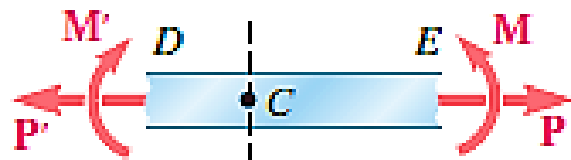


Fig. 4.39 (a) Member with eccentric loading. (b) Free-body diagram of the member with internal forces at section C.

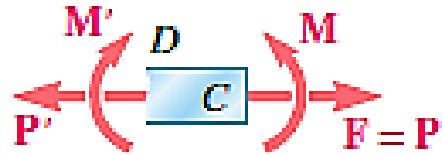


Photo 4.5 Walkway light.

δ_{avg}
+
 δ_{bend} $\Rightarrow$



(a)



(b)

Fig. 4.40 (a) Free-body diagram of straight portion DE. (b) Free-body diagram of portion CD.

Thus the stress distribution due to the original eccentric loading can be obtained by superposing the uniform stress distribution corresponding to the centric loads

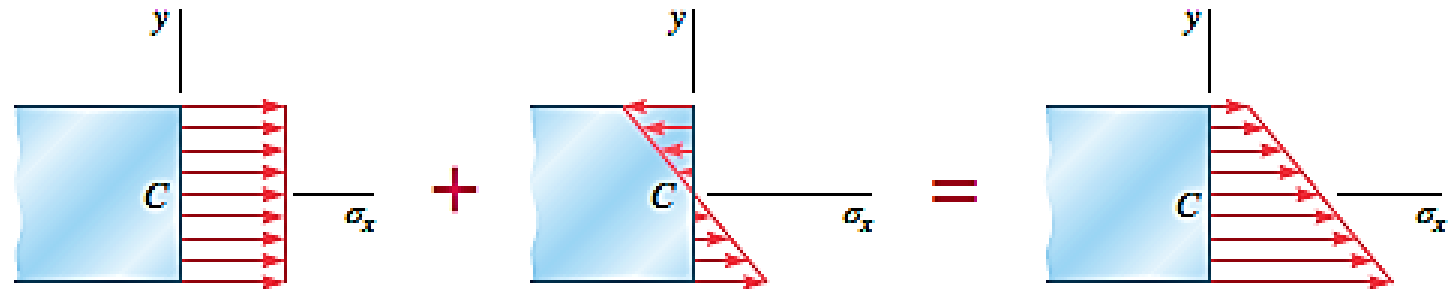


Fig. 4.41 Stress distribution for eccentric loading is obtained by superposing the axial and pure bending distributions.

$$\sigma_x = (\sigma_x)_{\text{centric}} + (\sigma_x)_{\text{bending}}$$

$$\sigma_x = \frac{P}{A} - \frac{My}{I}$$

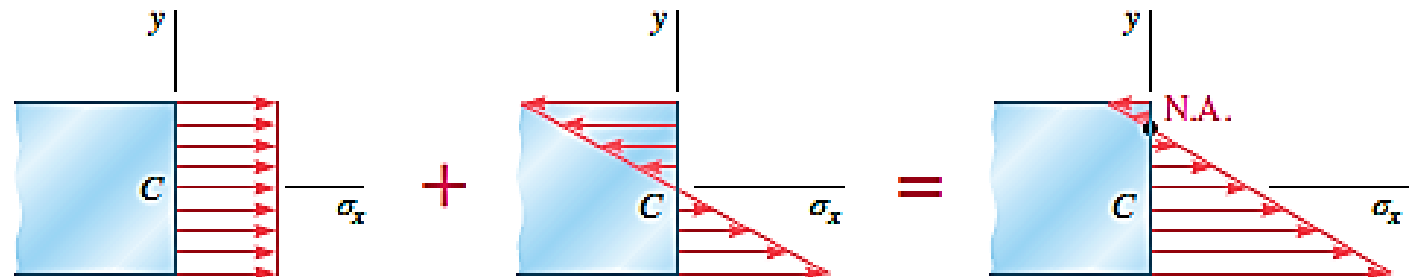


Fig. 4.42 Alternative stress distribution for eccentric loading that results in zones of tension and compression.

- Depending upon the geometry of the cross section and the eccentricity of the load, the combined stresses may all have the same sign.
- some may be positive and others negative. there will be a line in the section, along which $\sigma_x = 0$. This line represents the *neutral axis* of the section.

Example : Chain made by steel rod dia=12 mm. Determine the largest tensile and compressive stresses in the straight portion of a link, the distance between the centroidal and the neutral axis of a cross section

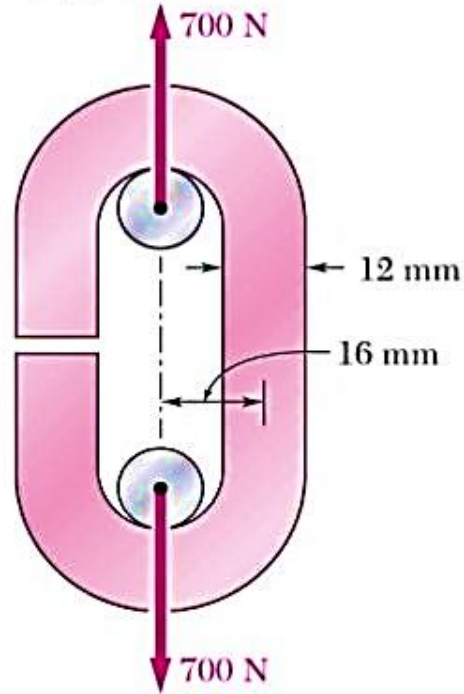


Fig. 4.52

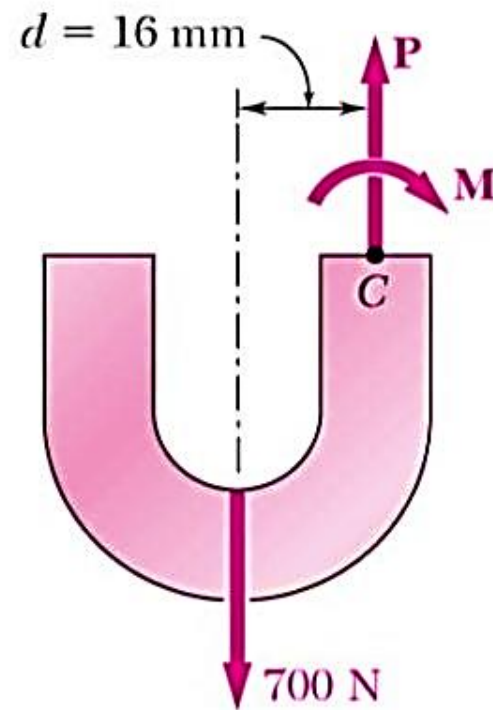


Fig. 4.53

$$P = 700 \text{ N}$$

$$M = Pd = 11.2 \text{ N.m}$$

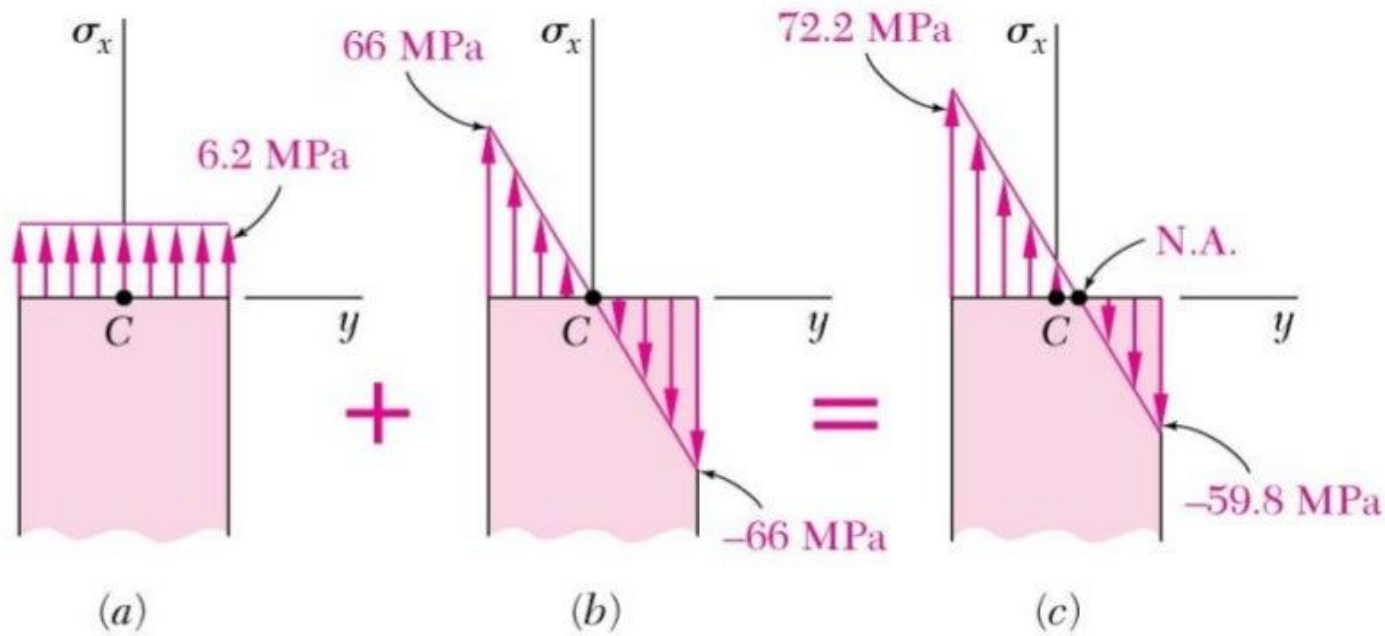


Fig. 4.54

$$\sigma_x = (\sigma_x)_{centric} + (\sigma_x)_{bending}$$

$$\sigma_x = \frac{P}{A} - \frac{My}{I}$$

$$A = \pi c^2 = \pi (6 \text{ mm})^2$$

$$I = \frac{1}{4} \pi c^4$$

-distance between centroidal and neutral axis

$$0 = \frac{P}{A} - \frac{M y_o}{I}$$

$$y_o = \left(\frac{P}{A} \right) \left(\frac{I}{M} \right)$$



Thank you

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